# Synthetic PET Image Generation Using GAN and cGAN for Data Augmentation

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## Problematic

Deep learning performance depends on large medical datasets, but PET[1] data is limited by acquisition cost, privacy constraints, and expert annotation, leading to overfitting, bias, and poor generalization.

[1] positron emission tomography

Growing use of deep learning in medical Imaging

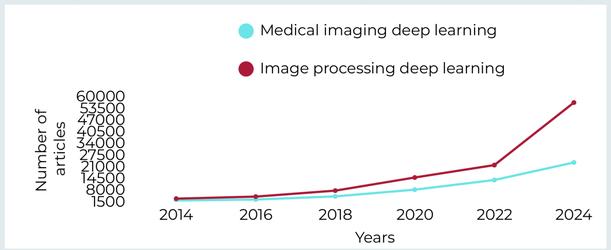
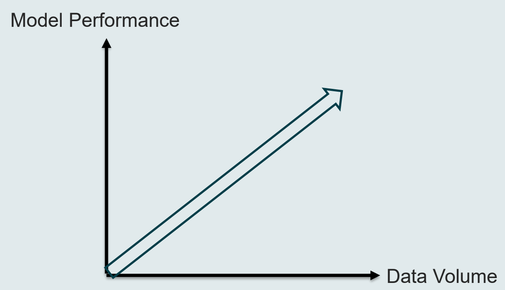
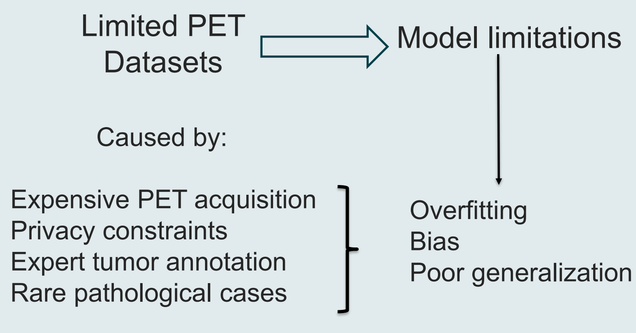



Figure 1: Growing use of deep learning in medical imaging. ScienceDirect.com | Science, health and medical journals, full text articles and books.

More training data can improve model performance.




## Objective

To investigate synthetic PET image generation as a data augmentation strategy and evaluate the suitability of generative models for producing structurally consistent PET images under limited-data conditions.

## State of the Art

Traditional Data Augmentation

Rotation / flipping
Cropping / scaling
Intensity transformation

Limitations

visual realism ≠ tumor-structure consistency

Gap persists

Related direction:

Conditional generation in medical imaging

GAN[2]-based Synthetic

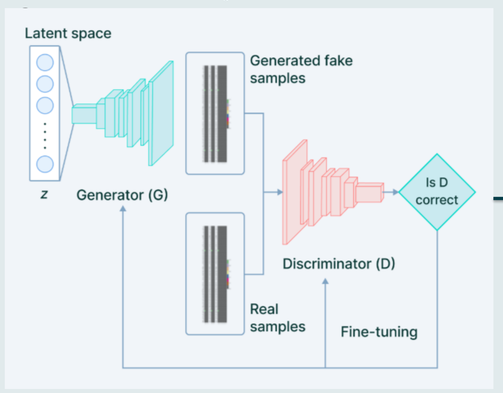



Figure 2: general GAN architecture
The Essential Guide to Neural Network Architectures

[2] Generative adversarial network

## Methodology

Data: 33 patients → 676 valid 2D slices after mask filtering.

The methodology compares several strategies for PET image generation.
A standard GAN was first used as a baseline.

Then, cGAN[3] models were implemented to guide generation using medical conditions.
Three conditions were compared:
**mask only, CT[4] only, and CT + mask**

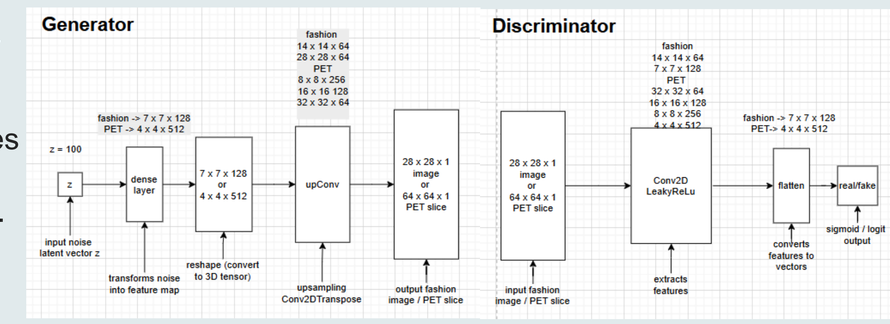



Figure 3: PET GAN architecture

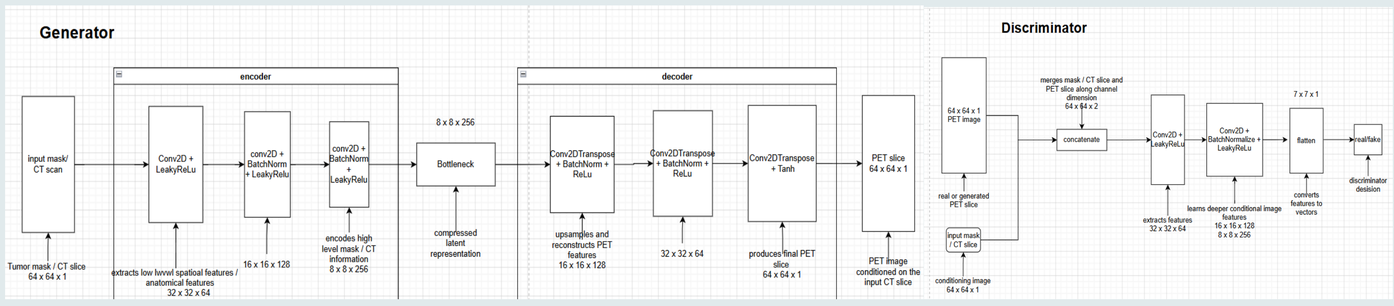



Figure 4: PET cGAN with mask-only / CT-only condition generator architecture

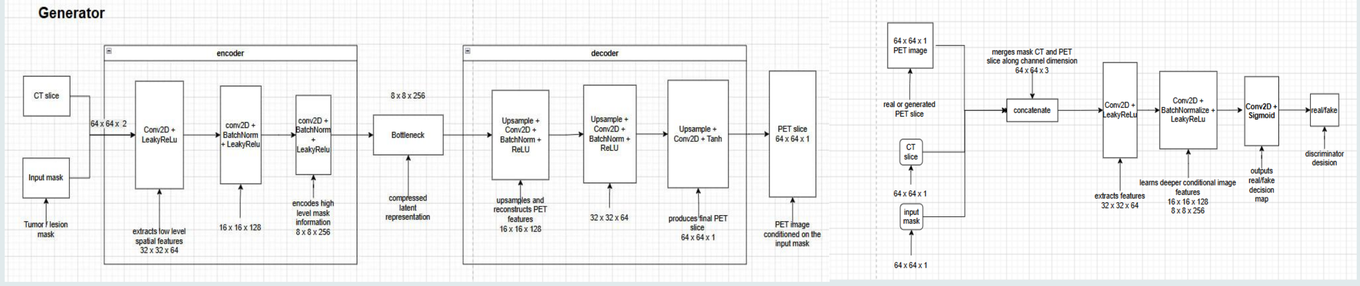



Figure 5: PET cGAN with mask and CT condition architecture

[3] Computed tomography
[4] Conditional GAN

## Results

GAN generation

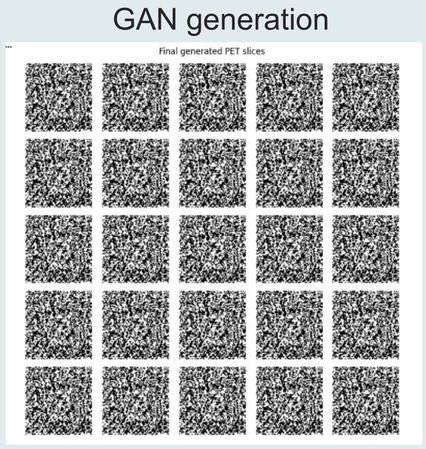



Figure 6: PET GAN output

cGAN generation with CT

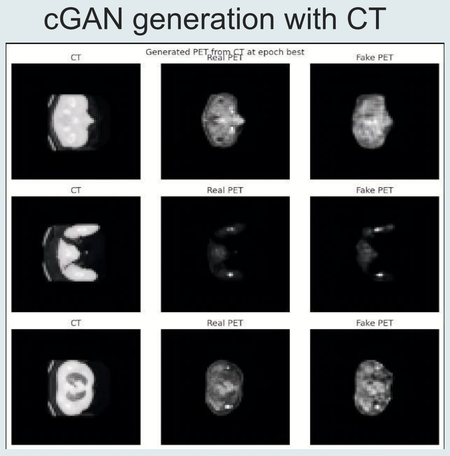



Figure 8: PET cGAN with CT-only conditioning

Image resize ratio and minimum mask pixels experiments outcomes

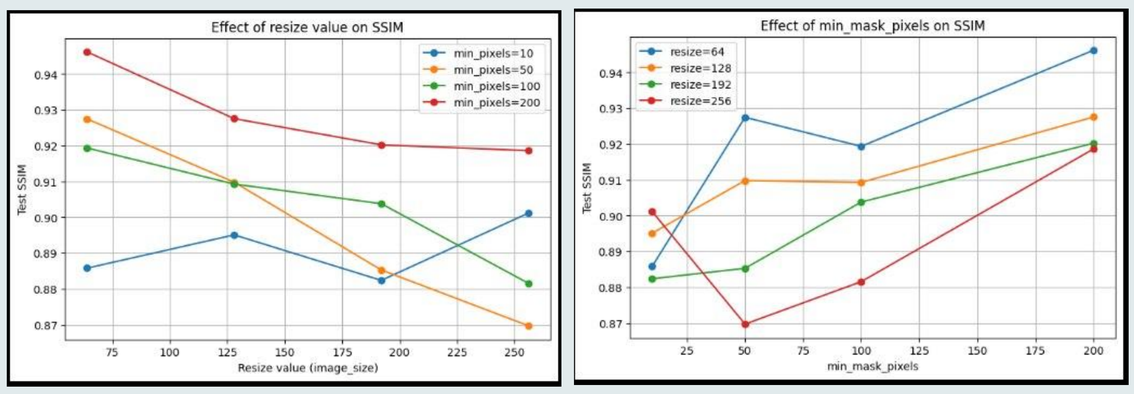



Figure 13: Experiment results(SSIM)

cGAN generation with mask

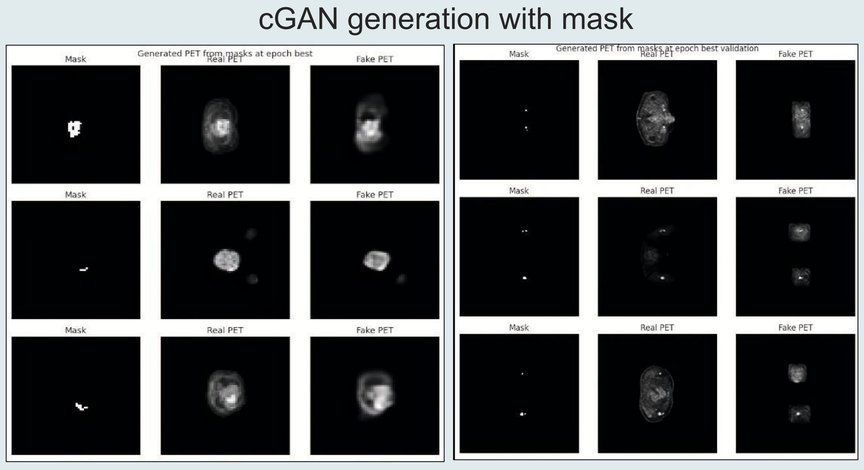



Figure 7: PET cGAN with mask-only conditioning - 2 cases

cGAN generation with CT and mask

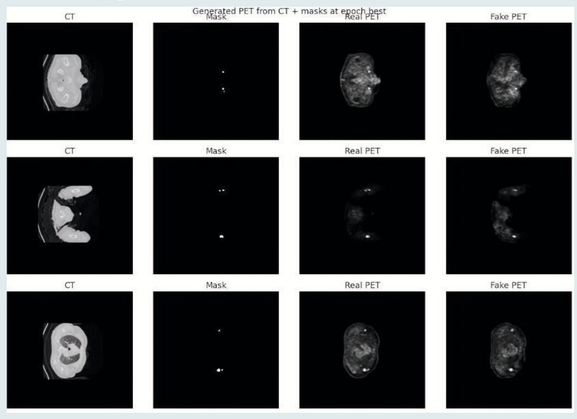



Figure 9: PET CT + mask cGAN results

CT-only achieved the highest PSNR/SSIM, while CT + mask was more medically meaningful because it combines anatomical context with lesion localization.

| Condition | PSNR[5] | SSIM[6] |
|---|---|---|
| Mask | 36.94 | 0.945 |
| CT | 37.01 | 0.963 |
| CT + Mask | 35.30 | 0.961 |

[5] Peak Signal to Noise Ratio
[6] Structure similarity

## Discussion

Conclusion:
- Standard GAN outputs lacked meaningful PET structure.
- Mask and CT conditioning improved generation quality.
- CT + mask gave the most coherent PET-like outputs.

Future work:
- Test more stable GAN variants, such as WGAN-GP[7] or pix2pix-style cGAN, to improve training stability.
- Extend evaluation with more PET/CT cases and expert visual assessment to confirm medical relevance.
- Compare segmentation results when training with **real PET only** versus **real + synthetic PET** data.

[7] Wasserstein GAN

## References


- Sherwani, M. K., & Gopalakrishnan, S. (2024). *A systematic literature review: deep learning techniques for synthetic medical image generation and their applications in radiotherapy.*
- Islam, J., & Zhang, Y. (2020). *GAN-based synthetic brain PET image generation.*
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., et al. (2014). *Generative adversarial nets.*


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More references